

41.6 Sensitivity analysis

Everything in the last three sections rested on missingness at random, and Proposition 41.1.1 says the data cannot check it. That is a reason neither to despair nor to pretend, but to report how much the conclusion depends on the assumption — which is what a sensitivity analysis does.

41.6.1 Two factorizations

The selection factorization Eq. 41.1.1 splits the joint law into a model for the data and a model for who responds. The other order gives the **pattern-mixture** factorization

$$f(\mathbf{D}, \mathbf{R} \mid \boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = f(\mathbf{D} \mid \mathbf{R}, \boldsymbol{\theta}_1) f(\mathbf{R} \mid \boldsymbol{\theta}_2), \quad (41.6.1)$$

a mixture, over patterns, of a different distribution of the data within each pattern. The two describe the same object and differ only in what is easy to say.

Proposition 41.6.1. Selection and pattern-mixture models

Consider a single variable v recorded when $R = 1$, alongside always recorded variables $\mathbf{u}$.

- (a) Every selection model $f(v \mid \mathbf{u}) \Pr(R = 1 \mid v, \mathbf{u})$ corresponds to the pattern-mixture model with

$$\begin{aligned} f(v \mid \mathbf{u}, R = r) &\propto f(v \mid \mathbf{u}) \Pr(R = r \mid v, \mathbf{u}), \\ \Pr(R = r \mid \mathbf{u}) &= \int f(v \mid \mathbf{u}) \Pr(R = r \mid v, \mathbf{u}) dv, \end{aligned}$$

and conversely, so neither formulation is more general.

- (b) The observed data identify $f(v \mid \mathbf{u}, R = 1)$ and $\Pr(R = 1 \mid \mathbf{u})$, and nothing about $f(v \mid \mathbf{u}, R = 0)$. The marginal

$$\begin{aligned} f(v \mid \mathbf{u}) &= \Pr(R = 1 \mid \mathbf{u}) f(v \mid \mathbf{u}, R = 1) \\ &\quad + \Pr(R = 0 \mid \mathbf{u}) f(v \mid \mathbf{u}, R = 0) \end{aligned} \quad (41.6.2)$$

is therefore identified only after the analyst supplies the last factor. Missing at random is the supply $f(v \mid \mathbf{u}, R = 0) = f(v \mid \mathbf{u}, R = 1)$.

- (c) Let the mechanism be $\Pr(R = 1 \mid v, \mathbf{u}) = h\{a(\mathbf{u}) + cv\}$, with h continuous, strictly increasing and valued in $(0, 1)$ and $a(\cdot)$ an unrestricted function. Then c is not identified: for every c at which the tilt Eq. 41.6.3 below is integrable and its integral covers the required value, there are an $a(\cdot)$ and an $f(v \mid \mathbf{u}, R = 0)$ reproducing the observed-data distribution exactly. Any standard error reported for c comes from restricting $a(\cdot)$ — to $a + \mathbf{u}^\top \mathbf{b}$, say — and from the assumed forms of h and $f(v \mid \mathbf{u})$, not from the data.

Proof.

- (a) Both expressions are the joint density $f(v, R = r \mid \mathbf{u})$ written in the two orders, by the definition of conditional probability; the displayed formulas are Bayes's rule in each direction, and the correspondence is one-to-one wherever the denominators are positive.
- (b) The observed data consist of $(\mathbf{u}, R)$ for all units and v for those with $R = 1$. Its distribution is determined by the law of $\mathbf{u}$, $\Pr(R = 1 \mid \mathbf{u})$ and $f(v \mid \mathbf{u}, R = 1)$, and

by nothing else, exactly as in Proposition 41.1.1. Equation Eq. 41.6.2 is the law of total probability. Setting the unidentified factor equal to the identified one is the MAR choice, since then $\Pr(R = 1 \mid v, \mathbf{u})$ computed by Bayes's rule is free of v .

- (c) Fix c , write $\pi(\mathbf{u}) = \Pr(R = 1 \mid \mathbf{u})$ and $f_1(v \mid \mathbf{u}) = f(v \mid \mathbf{u}, R = 1)$ for the two identified pieces, and abbreviate $h_c(v) = h\{a(\mathbf{u}) + cv\}$. Bayes's rule applied to (a) forces the unrecorded conditional to be the recorded one *tilted* by the odds of not responding:

$$f(v \mid \mathbf{u}, R = 0) = \frac{f_1(v \mid \mathbf{u})}{K(\mathbf{u})} \cdot \frac{1 - h_c(v)}{h_c(v)}, \quad K(\mathbf{u}) = \int f_1 \frac{1 - h_c}{h_c} dv. \quad (41.6.3)$$

Take that as the definition and assemble $f(v \mid \mathbf{u})$ from Eq. 41.6.2. Then $f(v \mid \mathbf{u})h_c(v) = \pi f_1 h_c + \{(1 - \pi)/K\}f_1(1 - h_c)$, which collapses to $\pi(\mathbf{u})f_1(v \mid \mathbf{u})$ — free of h_c — exactly when $K(\mathbf{u}) = \{1 - \pi(\mathbf{u})\}/\pi(\mathbf{u})$. Integrating over v then returns $\Pr(R = 1 \mid \mathbf{u}) = \pi(\mathbf{u})$, and dividing returns $f(v \mid \mathbf{u}, R = 1) = f_1(v \mid \mathbf{u})$: both identified pieces are reproduced, for this c .

It remains to solve $K(\mathbf{u}) = \{1 - \pi(\mathbf{u})\}/\pi(\mathbf{u})$ for the intercept $a(\mathbf{u})$, one scalar equation per $\mathbf{u}$. Since $(1 - h_c)/h_c$ decreases pointwise in $a(\mathbf{u})$, K is continuous and strictly decreasing, with $K \rightarrow 0$ as $a(\mathbf{u}) \rightarrow \infty$; a solution exists whenever the target lies in its range. Both provisos bite. For $h = \Phi$ and f_1 normal with standard deviation s , the integrand's logarithm behaves like $(c^2 - s^{-2})v^2/2$ as $v \rightarrow -\infty$, so K is finite exactly when $|c|s < 1$, and then runs over all of $(0, \infty)$; for larger $|c|$ no such pattern-mixture model exists and the construction is unavailable. Within the range, the observed-data likelihood profiled over the remaining parameters is constant in c , so c is not identified. Restricting $a(\mathbf{u})$ to $a + \mathbf{u}^\top \mathbf{b}$ leaves one equation per $\mathbf{u}$ and only $1 + \dim \mathbf{u}$ unknowns; the profile is then no longer flat, and the curvature — the reported standard error — comes from that restriction, not from the data. ■

Part (c) is worth dwelling on. Fit a Heckman-style selection model — a normal outcome equation and a probit selection equation, as in Heckman (1979) — and the software returns an estimate of the selection coefficient with a standard error; part (c) says that number is an artefact of the assumed functional forms. With an **exclusion restriction**, a variable entering the selection equation and not the outcome equation, the parameter becomes identified by more than functional form; without one it does not. Reporting such a fit as though the data had spoken is the central dishonesty of MNAR modelling, and the reason this section prefers to display the untestable part rather than estimate it.

Idea. Under MNAR nothing is estimated that was not assumed. The useful question is not “what is the answer?” but “how far must the assumption move before the answer changes?”

41.6.2 Delta-adjustment and tipping points

The pattern-mixture form suggests an honest device. Impute under MAR, then shift the imputed values by a fixed amount δ before analysing them: that is, assume

$$E(v \mid \mathbf{u}, R = 0) = E(v \mid \mathbf{u}, R = 1) - \delta. \quad (41.6.4)$$

Here δ is a *sensitivity parameter*: not estimated, but set. Zero is MAR, and a positive δ says the unrecorded values are worse than their recorded counterparts with the same covariates. Because Eq. 41.6.4 constrains only the unidentified factor of Eq. 41.6.2, every δ fits the observed records

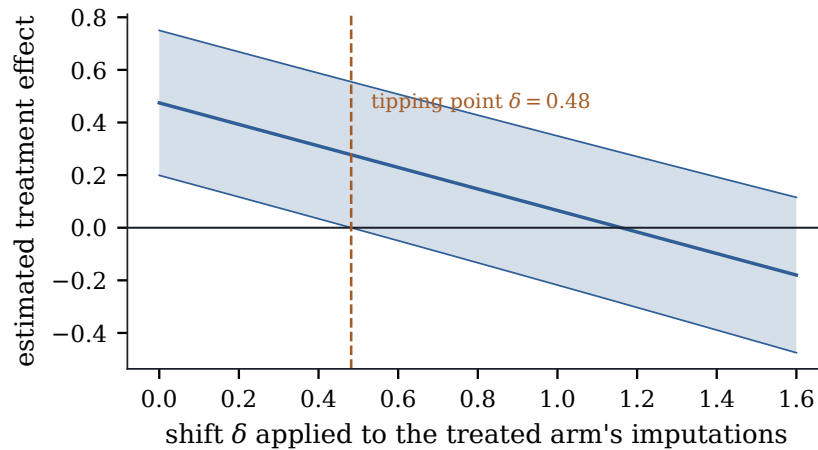


Figure 41.6.1: Estimated treatment effect and its 95% interval as the imputed outcomes of the treated arm are shifted down by δ . Every δ fits the recorded outcomes equally well. The tipping point is where the lower limit reaches zero.

equally well — in the example below all give the observed-data log-likelihood -373.62 — so the choice is a statement about the world, not about the data. Repeating the analysis over a grid of δ and reporting the curve is a **tipping-point analysis**: the value δ^* at which the conclusion reverses is the smallest departure from MAR that would overturn it.

Example. How far must the assumption move?

A two-arm study of 400 units records a baseline measurement for everyone and an outcome for those who stay. Dropout is likelier in the treated arm and at high baseline values, and 147 outcomes are lost — 83 treated and 64 control. Multiple imputation under MAR gives a treatment effect of 0.475 with 95% interval (0.199, 0.750).

Now suppose the treated units who dropped out would have done worse than the MAR imputation says, by δ . The estimate falls steadily and the lower limit reaches zero at $\delta^* = 0.482$, which is 0.452 of a residual standard deviation. The finding therefore survives the assumption that treated dropouts were up to about half a standard deviation worse than comparable stayers, and not more. Whether that is reassuring is a question for whoever knows why people dropped out.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(4108)
n, tau_true = 400, 0.6
arm = np.repeat([0, 1], n // 2)           # 0 = control, 1 = treated
base = rng.normal(size=n)                 # a baseline measurement,
                                         # never missing
y = 0.5 + tau_true * arm + 0.6 * base + rng.normal(size=n)
eta = -1.0 + 0.7 * arm + 0.8 * base       # dropout: depends on arm and
                                         # baseline
seen = rng.uniform(size=n) > 1 / (1 + np.exp(-eta))
print(f"{n - seen.sum()} of {n} outcomes lost:"
      f" {int((~seen & (arm == 1)).sum())} treated, {int((~seen & (arm == 0)).sum())} control")
```

[Run this code online](#)

```
Z = np.column_stack([np.ones(n), arm, base])

def mi_estimate(delta, M, rng):
    """Multiple imputation under MAR, then shift the treated arm's imputations by -
    delta."""
    Zo, k = Z[seen], int(seen.sum())
    g, *_ = np.linalg.lstsq(Zo, y[seen], rcond=None)
    sse = float(np.sum((y[seen] - Zo @ g) ** 2))
    qs, us = [], []
    for _ in range(M):
        s2 = sse / rng.chisquare(k - Z.shape[1])
        g_star = rng.multivariate_normal(g, s2 * np.linalg.inv(Zo.T @ Zo))
        draw = Z @ g_star + np.sqrt(s2) * rng.normal(size=n) - delta * arm
        y_full = np.where(seen, y, draw)
        b, *_ = np.linalg.lstsq(Z, y_full, rcond=None)
        resid = y_full - Z @ b
        v = (resid @ resid / (n - 3)) * np.linalg.inv(Z.T @ Z)[1, 1]
        qs.append(b[1])
        us.append(v)
    M_ = len(qs)
    q_bar, u_bar, b_var = np.mean(qs), np.mean(us), np.var(qs, ddof=1)
    total = u_bar + (1 + 1 / M_) * b_var
    gamma = (1 + 1 / M_) * b_var / total
    return q_bar, total, (M_ - 1) / gamma**2

deltas = np.linspace(0.0, 1.6, 33)
est, lo, hi = [], [], []
for d in deltas:
    q, t, nu = mi_estimate(d, 50, np.random.default_rng(4109))
    half = stats.t.ppf(0.975, nu) * np.sqrt(t)
    est.append(q)
    lo.append(q - half)
    hi.append(q + half)
est, lo, hi = map(np.array, (est, lo, hi))
tipping = float(np.interp(0.0, -lo, deltas))          # where the lower limit
reaches zero
print(f"delta = 0: effect {est[0]:.3f}, 95% interval ({lo[0]:.3f}, {hi[0]:.3f})")
print(f"tipping point: delta = {tipping:.3f}")
```

[Run this code online](#)

Variations abound. The shift may be applied to one arm or both, on the response scale or a link scale, and may depend on covariates or grow with time since dropout; in trials it is often replaced by **jump to reference**, in which a treated dropout is imputed as though from the control arm, or **copy increments in reference**, in which only the subsequent changes are borrowed. Each is a particular choice of the unidentified factor in [Eq. 41.6.2](#).

41.6.3 Honest reporting

The technical content of this section is thin; the discipline is not. A report should say how much is missing and where, by variable and by pattern; why, as far as the reasons recorded at collection time reveal, those being the only evidence about the mechanism that will ever exist; what was assumed, naming the variables MAR is conditional on, since “missing values were

imputed” states no assumption at all; what the method was, including whether the imputation model contained the response; and how far the conclusion travels, as a sensitivity curve or at least a tipping point. That last item is where regulators have converged: the National Research Council (2010) report recommends designing to prevent missingness and pre-specifying both the primary analysis and the sensitivity analyses, rather than choosing them after seeing the result.

Warning. A sensitivity analysis chosen after seeing the primary result is not a sensitivity analysis; it is a selection procedure, with the properties [Chapter 29](#) describes (Proposition 29.5.1). Fix the grid of δ and the summary to be reported before the analysis is run.

41.6.4 Exercises

41.6.4.1 A. Check your understanding

Exercise. A1

Write the bivariate normal of Exercise (B3) in both factorizations, selection and pattern-mixture, and identify which parameters are estimable from the observed data in each. Why is the pattern-mixture form better suited to a sensitivity analysis?

Exercise. A2

In [Example \(How far must the assumption move?\)](#) the tipping point is 0.482 on the outcome scale. Express it as a multiple of the estimated treatment effect and as a multiple of the residual standard deviation, and say which you would put in a report and why.

41.6.4.2 B. Practice

Exercise. B1

For the model of [Example \(How far must the assumption move?\)](#), show analytically that the treatment effect estimated after a δ -shift of the treated arm’s imputations is $\hat{\tau}(\delta) = \hat{\tau}(0) - \delta p_1$, where p_1 is the fraction of treated units whose outcome is missing, provided the analysis model contains the arm indicator. Check the slope against the figure.

Solution. With the arm indicator in the model and a completed data set, the estimated effect is a difference of arm means adjusted for the baseline. Shifting the imputed outcomes of the treated arm down by δ lowers that arm’s adjusted mean by δ times the fraction of its units that were imputed, and leaves the control arm alone; the baseline adjustment is unaffected because the shift is constant within the arm. Hence $\hat{\tau}(\delta) = \hat{\tau}(0) - \delta p_1$. With 83 of 200 treated units missing, $p_1 \approx 0.42$, and the slope of the curve in [Figure 41.6.1](#) matches.

41.6.4.3 C. Going deeper

Exercise. C1

For a monotone pattern with three time points, state the **complete-case missing value**

restriction — the unidentified conditional for a unit that dropped out after time j equals the corresponding conditional among units seen at all three times — and the **neighbouring-case** alternative. Show that both are particular choices of the unidentified factor in [Eq. 41.6.2](#), and that the MAR choice lies between them in a sense you should make precise.

Exercise. C2

Suppose the response is bounded in $[0, 1]$. Compute the **worst-case bounds** for its mean: impute every missing value by 0, then by 1. Show that the width of the resulting interval is the fraction missing, that it requires no assumption at all, and that it is usually too wide to be useful. Discuss where such bounds nevertheless earn their place.

Solution. If a fraction λ of values is missing, the mean of the completed data ranges over an interval of width λ as the imputations range over $[0, 1]$, and every value in that interval is attained by some choice, so the bounds are sharp and assumption-free — they are the Manski-style bounds for a bounded outcome. They are useless when λ is large compared with the effect of interest, which is the usual case. They earn their place as a first line in a report: if the worst-case bound already excludes the null, no assumption about the mechanism is needed at all, and that is worth knowing before any modelling begins.